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Vision Transformer-Based Boat Detection System: A Deep Learning Approach

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Abstract

Vision Transformers (ViTs) have revolutionized the field of computer vision by offering an alternative to convolutional neural networks (CNNs) for various tasks including image classification, object detection, and semantic segmentation. This paper explores the design and implementation of a Vision Transformer-based boat detection system, aimed at enhancing maritime surveillance, port management, and environmental monitoring. We compare the performance of ViT models with CNN-based approaches, discuss datasets, training methodologies, and propose a framework suitable for real-time deployment in coastal and satellite imagery scenarios.

Keywords: Vision Transformer, Boat Detection, Maritime Surveillance, Deep Learning, Object Detection, Satellite Imagery

1. Introduction

Maritime surveillance is critical for ensuring navigational safety, detecting illegal activities, and supporting maritime rescue operations. Traditional methods relying on radar and manual observation are often limited by weather conditions and human error. With the advent of high-resolution aerial and satellite imagery, automated boat detection has become feasible. This paper investigates the application of Vision Transformers (ViTs) for this task, leveraging their capacity to model long-range dependencies and capture contextual information.

2. Background and Related Work

Object detection in computer vision has historically been dominated by CNNs, with models like Faster R-CNN, YOLO, and SSD achieving remarkable success. More recently, transformers—originally developed for natural language processing—have shown promise in vision tasks through self-attention mechanisms. ViTs process images as sequences of patches, enabling flexible global feature aggregation. This paradigm has demonstrated superior performance on tasks like image classification (ViT, DeiT), detection (DETR), and segmentation (SegFormer).

3. Vision Transformers for Object Detection

3.1. Architecture Overview

ViTs divide images into fixed-size patches, flatten them, and embed them into a sequence of tokens that are processed through multiple transformer layers. Key components include multi-head self-attention, positional encoding, and MLP blocks.

3.2. DETR: Detection Transformer

DETR reframes object detection as a direct set prediction problem, using a transformer encoder-decoder architecture. It eliminates the need for hand-crafted anchors and non-maximum suppression.

3.3. Advantages Over CNNs

ViTs provide better global context understanding and are less prone to overfitting on small datasets when pre-trained effectively. They also show robustness in cluttered or low-resolution scenarios, common in satellite imagery.

4. System Design and Implementation

4.1. Dataset and Preprocessing

We utilize public datasets such as SeaDronesSee, Airbus Ship Detection Challenge, and satellite imagery annotated for maritime vessels. Data augmentation techniques including flipping, rotation, and color jittering are applied.

4.2. Model Configuration

The system employs a DETR model pre-trained on COCO and fine-tuned for boat detection. Hyperparameters like learning rate, weight decay, and number of heads are optimized.

4.3. Training Pipeline

The training is conducted on GPUs using the PyTorch framework. Early stopping, learning rate scheduling, and checkpointing are integrated for efficient training.

4.4. Inference and Post-Processing

During inference, the model outputs bounding boxes and class labels. Post-processing involves thresholding confidence scores and optional tracking using Kalman filters.

5. Experimental Results

5.1. Evaluation Metrics

We use precision, recall, mean average precision (mAP), and inference time to evaluate model performance.

5.2. Comparative Analysis

Our ViT-based model outperforms traditional CNN detectors on key metrics, especially in scenarios with occlusions and varying illumination.

5.3. Visualization

Detection outputs are visualized on test images. The ViT model effectively distinguishes boats from other maritime structures.

6. Applications and Use Cases

Applications of this system include coast guard surveillance, harbor traffic monitoring, marine pollution detection, and illegal fishing control. The flexibility of ViTs enables adaptation to various sensor modalities and geographic regions.

7. Challenges and Limitations

ViT models are computationally intensive and require substantial training data. In maritime environments, variable weather, wave patterns, and object scales pose additional challenges. Data scarcity and labeling costs remain barriers to generalization.

8. Future Work

We plan to explore lightweight ViT variants like MobileViT for edge deployment, integrate temporal information for video-based detection, and enhance domain adaptation for transfer to new maritime regions.

9. Conclusion

Vision Transformers hold significant potential for advancing boat detection systems, offering improved accuracy and robustness over CNN counterparts. Their adoption in real-world applications can enhance maritime situational awareness and contribute to global marine safety efforts.

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